

NGUYEN GIA HUY

FinTech Backend Engineer – Core Banking

johnnynguyen882@gmail.com | 0948385149 | linkedin.com/in/huy8bit

PROFESSIONAL SUMMARY

Backend engineer at Vikki Digital Bank, working on the Lending and Overdraft products built on Thought Machine Vault Core. Experience covers interest and fee calculation, repayment and settlement, account migration, and integration with surrounding banking systems. Previously worked on multi-tenant security for VNPAY's fintech cloud platform.

TECHNICAL SKILLS

Languages: Python, Golang, SQL
Core Banking: Thought Machine Vault Core, Python Smart Contracts
Backend & Data: gRPC, REST APIs, PostgreSQL, Redis, Kafka
Cloud: AWS (EKS, ECS, Lambda, RDS), Kubernetes, Docker
Security: Keycloak, RBAC, Kong Gateway

WORK EXPERIENCE

Middle Software Engineer, Core Banking

03/2026 – Present

Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Developed and maintained the bank's Lending and Overdraft products as Python Smart Contracts on Thought Machine Vault Core.
- Implemented interest accrual for both products, scheduled through Vault Core events.
- Implemented penalty and late-fee calculation for overdue payments.
- Implemented repayment and settlement logic, which applies customer payments and closes out the amounts owed under each product.
- Integrated Vault Core with the bank's external backend services for core banking operations and transaction workflows.
- Built the account migration system in Golang on AWS ECS, Lambda, and RDS, reducing migration time from **2.5 hours to 27 minutes**.
- Rebuilt the product simulation testing framework in Golang, replacing Gauge, so that contract behaviour can be verified in **just over 50 seconds instead of 24 minutes**.
- Deployed a new Vault Core version on AWS (EKS, RDS, ECR, ELB) and built the CI/CD pipeline for automated build, test, and deployment.

Software Engineer

04/2025 – 03/2026

Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Built and operated ETL pipelines on AWS (Lambda, ECS, S3, Athena, RDS) for au Denki's large-scale utility data.
- Managed the pipeline infrastructure with Terraform and AWS SAM, with monitoring in CloudWatch and DataDog.
- Raised batch throughput by **150%** with a pipeline redesign that added parallel execution and more efficient data access.
- Migrated the search and analytics platform to OpenSearch and redesigned its ingestion and indexing.

Software Engineer – Fintech Cloud Platform

04/2024 – 04/2025

VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Implemented multi-tenant authentication and authorization with Keycloak and RBAC for **20+** fintech and enterprise customer organizations, keeping each tenant's access isolated.
- Enforced service-level and network-level access control for **100+** microservices through Kong Gateway.
- Operated VNPAY Cloud's OpenStack infrastructure (**50+** production Kubernetes clusters), where Cluster API handled provisioning, scaling, and zero-downtime upgrades.
- Automated roughly **80%** of infrastructure provisioning and configuration in the data center.

Blockchain Developer – DeFi & Transaction Systems

06/2023 – 09/2023

EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Developed and reviewed **15+** smart contracts for DeFi and NFT use cases in Solidity and Move.
- Integrated backend services with **5+** blockchain networks (Ethereum, BSC, Polygon, Aptos, Sui), keeping transaction handling consistent and off-chain records in line with on-chain data.
- Built event-driven services that sync on-chain events into off-chain databases and APIs.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Built a backend on gRPC-Gateway, Kafka, MongoDB, and Redis that handled up to **1.5K** requests per second at 80–100 ms latency, with multi-cluster failover and observability for **99.95%** uptime.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. in Information Technology, Knowledge Engineering

DevOps on AWS Specialization | Coursera